

Homework Assignment - PageRank

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In this homework we discuss the PageRank algorithm as detailed in the paper "*The \$25,000,000,000 Eigenvector: The Linear Algebra Behind Google*".

The project is composed of four main components:

1. **The General Power Method:** Implementing the core iterative solver.
2. **The PageRank Algorithm:** Implementing the specific Google Matrix formulation $(1-m)A+mS$.
3. **Dataset Analysis:** Processing the web graph (`hollins.dat`) and handling dangling nodes.
4. **Theoretical Exercises:** Solutions to Exercises 4 and 10.

1 The General Power Method

1.1 Theoretical Background

As outlined in **Section 4** of the paper, the **Power Method** constitutes an iterative approach used to approximate the eigenvalue and eigenvector of a matrix.

Consider a column-stochastic matrix A (where each column sums to 1) the sequence defined iteratively is:

$$x_k = Ax_{k-1}$$

To maintain stability and monitor convergence we normalize the vector during each iteration. For a column-stochastic matrix corresponding to a web the principal eigenvalue is $\lambda_1 = 1$.

1.2 Implementation

The code employs simple data structures to execute the method. It utilizes the **Rayleigh Quotient** for approximating the eigenvalue during each iteration.

- **Input:** A square matrix A , an initial vector v , maximum iterations, and a fixed tolerance for convergence.
- **Process:** It calculates $v_{new} = Av$, normalizes this vector and approximates the eigenvalue λ via the Rayleigh Quotient ($\lambda = \frac{v^T Av}{v^T v}$).
- **Stopping Criterion:** The loop stops when the relative change in the estimated eigenvalue drops below the tolerance threshold.
- **Output:** The algorithm returns the estimated dominant eigenvalue (*lambda_new*) and the corresponding normalized eigenvector(*v*).

2 The Google PageRank Algorithm

2.1 Theoretical Background

As mentioned in **Section 3** ("A remedy for $\dim(V_1(A)) > 1$ "), the use of raw link matrix A has one big deficiency: if the web is not connected (it has disjoint sub-webs), the ranking vector is not unique. To solve this, we replace A with the **Google Matrix** M :

$$M = (1 - m)A + mS$$

where:

- m is the **damping factor** (usually 0.15).
- S is a matrix whose entries are all equal to $1/n$.

This makes M a **positive column-stochastic matrix**, and by **Theorem 4.2** it possesses a unique eigenvector q with eigenvalue 1, whose components are positive.

2.2 Computational Optimization

We do not want to actually form M since this is computationally expensive. Moreover, this would destroy the sparsity of A , converting a matrix with few non-zeros to one that is full of non-zero entries.

However, we can rewrite the product Mx efficiently as given by **Equation 3.2**:

$$x = (1 - m)Ax + ms$$

where s is a column vector whose all entries are $1/n$.

2.3 Implementation Details

We updated the code in order to use this optimized iterative process:

1. Begin with a guess vector x_{old} .
2. Calculate the new iteration by using the following formula: $x_{new} = (1 - m)Ax_{old} + ms$.
3. Repeat until the difference between iterations $\|x_{new} - x_{old}\|_1$ is less than our tolerance, conventionally set to $1e-6$.
4. Return ranked pages based on their score.

3 Test on Figures 2.1 and 2.2

We present the results obtained by running the implemented code on the graphs described in Figure 2.1 and Figure 2.2 of the paper.

3.1 Figure 2.1 (Connected Web)

The PageRank algorithm converged in **19 iterations**. The final rankings are detailed below.

Ranking results for Figure 2.1 (Connected Web)

Rank	Page	Score
1	Page 1	0.3682
2	Page 3	0.2880
3	Page 4	0.2021
4	Page 2	0.1418

3.2 Figure 2.2 (Disconnected Web)

The PageRank algorithm converged in **2 iterations**. The final rankings are detailed below.

Ranking results for Figure 2.2 (Disconnected Web)

Rank	Page	Score
1	Page 3	0.2850
2	Page 4	0.2850
3	Page 1	0.2000
4	Page 2	0.2000
5	Page 5	0.0300

The testing output on Figure 2.2 confirms the existence of a unique ranking (eigenvector) for the modified matrix M , guaranteed by damping factor $m=0.15$.

4 Processing the "Hollins" Dataset

4.1 Data Format Analysis

The given dataset `hollins.dat` contains both URL definitions and link structures. In order to correctly process it, the following steps are necessary:

1. **Parse metadata:** Read the total number of pages and links.
2. **Map Indices:** The file uses 1-based indexing while Python uses 0-based indexing. We have to adjust them.
3. **Construct Matrix A :** Formulate the transition probability matrix where $A_{ij} = 1/k$ if page j links to i and has k total outgoing links.

4.2 Handling Dangling Nodes

As pointed out in **Section 2.2.2** of the paper, pages with no outgoing links result in columns of all zeros in the link matrix. This makes the matrix column-substochastic, the sums being less than 1, hence violating the condition for **Proposition 1** that 1 should be an eigenvalue.

To fix this, and to make sure that our PageRank vector sums exactly to 1, we apply a **Stochastic Correction**:

- If a page j has no outgoing links (`out_degree[j] == 0`), we replace that zero-column with uniform probabilities ($1/n$). This implies that when a user hits a dead-end page, they jump to any other page in the web with equal probability.

We implemented a specific function `load_and_process_data()` which reads the file, handles the mixed format, applies the stochastic correction, and then we ran the PageRank algorithm defined in Step 2 on the output.

4.3 Results from Hollins Dataset

The algorithm was executed on the processed `hollins.dat` dataset. The process details and the top 10 rankings are presented below.

Reading file `hollins.dat`...

Parsed 6012 URLs and 23875 links.

Corrected 3189 dangling nodes (dead ends) by making them random jumps.

Calculating PageRank...

PageRank converged in 58 iterations.

Top 10 Ranked Webpages

Rank	Score	ID	URL
1	0.019879	2	http://www.hollins.edu/
2	0.009288	37	http://www.hollins.edu/admissions/visit/visit.htm
3	0.008610	38	http://www.hollins.edu/about/about_tour.htm
4	0.008065	61	http://www.hollins.edu/htdig/index.html
5	0.008027	52	http://www.hollins.edu/admissions/info-request/info-request.cfm
6	0.007165	43	http://www.hollins.edu/admissions/apply/apply.htm
7	0.006583	425	http://www.hollins.edu/academics/library/resources/web.linx.htm
8	0.005989	27	http://www.hollins.edu/admissions/admissions.htm
9	0.005572	28	http://www.hollins.edu/academics/academics.htm
10	0.004452	4023	http://www1.hollins.edu/faculty/saloveyca/clas%20395/Sculpture/sld001.htm

Verification check: Sum of all PageRank scores: 1.0 (which matches the theory)

The analysis of the site ****hollins.edu**** confirms that the algorithm identifies authoritative pages and characterizes the hierarchical web structure.

Dominant Authority: According to expectations, the homepage has the highest score (Rank 1, score 0.019879) and is over double the next page's score (0.009288), as it is the root domain.

High-Value Sections: Pages for Admissions, Visiting, and Core Academics are determined as major hubs, hence Ranks 2, 3, 5, 6, 8, considering the internal linking structure favors recruitment and primary institutional information.

Deep Links: Low scores for pages related to Library resources or some faculty slides prove that the algorithm provides less authority to deep and specific resources when compared with the pages of major hubs.

5 Exercises 4 and 10

5.1 Exercise 4

Problem: In the web of Figure 2.1, remove the link from page 3 to page 1. In the resulting web, page 3 is now a dangling node.

1. Set up the corresponding substochastic matrix.
2. Find its largest positive (Perron) eigenvalue.
3. Find a non-negative Perron eigenvector for this eigenvalue.
4. Does the resulting ranking seem reasonable?

Solution

- Page 1 links to 2, 3, 4 (3 links $\rightarrow$ weight $1/3$).
- Page 2 links to 3, 4 (2 links $\rightarrow$ weight $1/2$).
- Page 3 links to 1 (1 link $\rightarrow$ weight 1).
- Page 4 links to 1, 3 (2 links $\rightarrow$ weight $1/2$).

The original matrix is:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

Removing the link from Page 3 to Page 1 makes column 3 all zeros. The new substochastic matrix is:

$$A_{new} = \begin{bmatrix} 0 & 0 & \mathbf{0} & 1/2 \\ 1/3 & 0 & \mathbf{0} & 0 \\ 1/3 & 1/2 & \mathbf{0} & 1/2 \\ 1/3 & 1/2 & \mathbf{0} & 0 \end{bmatrix}$$

Finding the Eigenvalues

We solve $\det(A - \lambda I) = 0$. Expanding along the 3rd column (which has $-\lambda$ on the diagonal and zeros elsewhere):

$$\det(A - \lambda I) = (-\lambda) \cdot \det \begin{bmatrix} -\lambda & 0 & 1/2 \\ 1/3 & -\lambda & 0 \\ 1/3 & 1/2 & -\lambda \end{bmatrix}$$

(Note: The 3×3 minor is obtained by removing row 3 and col 3 from the original 4×4 system).

Calculating the determinant of the 3×3 matrix:

$$D_3 = -\lambda(\lambda^2 - 0) - 0 + \frac{1}{2} \left(\frac{1}{3} \cdot \frac{1}{2} - (-\lambda) \cdot \frac{1}{3} \right)$$

$$D_3 = -\lambda^3 + \frac{1}{2} \left(\frac{1}{6} + \frac{\lambda}{3} \right) = -\lambda^3 + \frac{\lambda}{6} + \frac{1}{12}$$

So the characteristic equation is:

$$-\lambda \left(-\lambda^3 + \frac{\lambda}{6} + \frac{1}{12} \right) = 0$$

$$\lambda^4 - \frac{\lambda^2}{6} - \frac{\lambda}{12} = 0 \quad \implies \quad \lambda(12\lambda^3 - 2\lambda - 1) = 0$$

We look for the largest positive root of $12\lambda^3 - 2\lambda - 1 = 0$. By numerical approximation:

$$\boxed{\lambda \approx 0.555}$$

Finding the Eigenvector

We solve $(A - \lambda I)x = 0$ with $\lambda \approx 0.555$.

$$\begin{aligned} 1) \quad & -\lambda x_1 + 0.5x_4 = 0 \implies x_4 = 2\lambda x_1 \\ 2) \quad & \frac{1}{3}x_1 - \lambda x_2 = 0 \implies x_2 = \frac{1}{3\lambda}x_1 \\ 3) \quad & \frac{1}{3}x_1 + \frac{1}{2}x_2 + 0.5x_4 = \lambda x_3 \implies x_3 = \frac{1}{\lambda}\left(\frac{x_1}{3} + \frac{x_2}{2} + \frac{x_4}{2}\right) \end{aligned}$$

Let $x_1 = 1$. Then:

- $x_1 = 1$
- $x_4 = 2(0.555) = 1.11$
- $x_2 = 1/(3 \times 0.555) \approx 0.60$
- $x_3 = \frac{1}{0.555}(0.333 + 0.30 + 0.555) \approx \frac{1.188}{0.555} \approx 2.14$

Vector $v \approx [1, 0.60, 2.14, 1.11]^T$. Sum ≈ 4.85 . Normalized Ranking:

$$x \approx \begin{bmatrix} 0.206 \\ 0.124 \\ 0.441 \\ 0.229 \end{bmatrix}$$

Is this reasonable?

The ranking places Page 3 (0.441) significantly higher than all others, despite Page 3 having no outgoing links. This confirms the "dangling node" problem: Page 3 acts as a rank sink, absorbing importance from the network without redistributing it. While the math is correct for the substochastic matrix, the ranking is **not reasonable** for a functional search engine, as it disproportionately rewards pages that do not contribute to the web's connectivity.

The Corrected Matrix

The paper mentions that dangling nodes (like Page 3) cause issues. The standard mathematical correction (often applied before using the damping factor m) is to replace the column of zeros with a column of $1/n$. This represents a "random jump": if a surfer reaches Page 3, they jump to any page with equal probability ($1/4$). We replace column 3 (zeros) with $[1/4, 1/4, 1/4, 1/4]^T$. The new matrix is column-stochastic (columns sum to 1):

$$\bar{A} = \begin{bmatrix} 0 & 0 & 1/4 & 1/2 \\ 1/3 & 0 & 1/4 & 0 \\ 1/3 & 1/2 & 1/4 & 1/2 \\ 1/3 & 1/2 & 1/4 & 0 \end{bmatrix}$$

5.2 Exercise 10

Problem: Suppose that A is the link matrix for a strongly connected web of n pages. Show that $\dim(V_1(A)) = 1$ by following the steps below.

Solution

1. Show that $(A^2)_{ij} > 0$ if and only if page i can be reached from page j in exactly two steps.

By the definition of matrix multiplication:

$$(A^2)_{ij} = \sum_{k=1}^n A_{ik} A_{kj}$$

Since A is a link matrix, all entries are non-negative ($A_{xy} \geq 0$). For the sum to be strictly positive ($(A^2)_{ij} > 0$), at least one term in the summation must be positive. This means there exists some intermediate page k such that $A_{ik} > 0$ and $A_{kj} > 0$.

In graph terms:

- $A_{kj} > 0$ implies a link $j \rightarrow k$.
- $A_{ik} > 0$ implies a link $k \rightarrow i$.

Therefore, a path $j \rightarrow k \rightarrow i$ exists, representing exactly two steps.

2. Show generally that $(A^p)_{ij} > 0$ if and only if page i can be reached from page j in exactly p steps.

We proceed by induction. The base case ($p = 1$) is the definition of the link matrix, and we proved $p = 2$ above.

Assume true for $p - 1$. Then:

$$(A^p)_{ij} = (A \cdot A^{p-1})_{ij} = \sum_{k=1}^n A_{ik} (A^{p-1})_{kj}$$

For $(A^p)_{ij} > 0$, there must be at least one k where $A_{ik} > 0$ (link $k \rightarrow i$) and $(A^{p-1})_{kj} > 0$ (path $j \rightarrow \dots \rightarrow k$ of length $p - 1$). Combining a path of length $p - 1$ ending at k with a single step $k \rightarrow i$ yields a path of exactly p steps.

3. Argue that $(I + A + A^2 + \dots + A^p)_{ij} > 0$ if and only if page i can be reached from page j in p or fewer steps.

Let $S_p = I + A + A^2 + \dots + A^p$. Since all matrices involved consist of non-negative entries, the entry $(S_p)_{ij}$ is positive if and only if the (i, j) entry of *at least one* term in the sum is positive.

$$(S_p)_{ij} > 0 \iff \exists k \in \{0, 1, \dots, p\} \text{ such that } (A^k)_{ij} > 0$$

Since $(A^k)_{ij} > 0$ corresponds to a path of exactly k steps (where $k = 0$ implies $i = j$), the sum being positive implies a path exists of length $0, 1, \dots$, or p .

4. Explain why $I + A + A^2 + \dots + A^{n-1}$ is a positive matrix if the web is strongly connected.

A web is **strongly connected** if there is a path from any node j to any node i . In a graph with n nodes, if a path exists, the shortest path between any two nodes must have length at most $n - 1$ (visiting every other node once).

Therefore, for every pair (i, j) , there exists some $k \leq n - 1$ such that page i is reachable from page j in k steps. By the previous step, this guarantees that the (i, j) entry of the sum $(I + \dots + A^{n-1})$ is strictly positive. Thus, the matrix is positive.

5. Use the last part to show that $B = \frac{1}{n}(I + A + A^2 + \dots + A^{n-1})$ is positive and column-stochastic.

- **Positive:** We just showed the sum is a positive matrix. Multiplying by $\frac{1}{n}$ keeps it positive.

- **Column-Stochastic:** The product of column-stochastic matrices is column-stochastic. Since A is column-stochastic, $A^2, A^3, \dots$ are all column-stochastic. The identity I is also column-stochastic. Let $\mathbf{1}$ be the column vector of all ones. For any stochastic matrix M , $\mathbf{1}^T M = \mathbf{1}^T$. Summing the columns of B :

$$\mathbf{1}^T B = \frac{1}{n} \mathbf{1}^T (I + A + \dots + A^{n-1}) = \frac{1}{n} (\mathbf{1}^T + \mathbf{1}^T + \dots + \mathbf{1}^T)$$

Since there are n terms in the sum:

$$\mathbf{1}^T B = \frac{1}{n} (n \cdot \mathbf{1}^T) = \mathbf{1}^T$$

Thus, the column sums of B are all 1.

By Lemma 3.2, since B is positive and column-stochastic, $\dim(V_1(B)) = 1$.

6. **Show that if $x \in V_1(A)$ then $x \in V_1(B)$. Why does this imply that $\dim(V_1(A)) = 1$?**

Let $x \in V_1(A)$. By definition, $Ax = 1x = x$. Iterating this, $A^2x = A(Ax) = Ax = x$. generally, $A^kx = x$ for any integer $k \geq 0$.

Now apply B to x :

$$\begin{aligned} Bx &= \frac{1}{n} (I + A + A^2 + \dots + A^{n-1})x \\ &= \frac{1}{n} (Ix + Ax + A^2x + \dots + A^{n-1}x) \\ &= \frac{1}{n} (x + x + x + \dots + x) \quad (n \text{ times}) \\ &= \frac{1}{n} (nx) \\ &= x \end{aligned}$$

Thus, $Bx = x$, which implies $x \in V_1(B)$.

This proves that $V_1(A) \subseteq V_1(B)$. Since we know from the previous step that $\dim(V_1(B)) = 1$, the subspace $V_1(A)$ must have dimension ≤ 1 . However, since A is column-stochastic, we know 1 is an eigenvalue (Proposition 1), so $\dim(V_1(A)) \geq 1$.

Therefore, $\dim(V_1(A)) = 1$.

6 Conclusions

The work demonstrated that while the ideal scenario (as proven by Exercise 10 for a strongly connected web) guarantees a unique and stable ranking ($\dim(V_1(A)) = 1$), the raw link structure of the web contains critical flaws. Specifically, the existence of dangling nodes (as explored in Exercise 4) necessitates an immediate correction to prevent the loss of rank mass and ensure the resulting matrix remains column-stochastic. Furthermore, to address issues like reducibility (disconnected components) and guarantee fast convergence across the entire web, the final model must incorporate the damping factor (m). Ultimately, the PageRank vector is successfully calculated only by using the Google Matrix (M) (an augmented, positive, and irreducible matrix) derived from the raw link structure; this mathematical concept ensures that the principles of the *Perron-Frobenius Theorem* are met, yielding a robust, unique, and meaningful measure of importance for every page in the graph.